

Nikhil Kotikalapudi

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EDUCATION

The Pennsylvania State University – Schreyer Honors College — School of EECS

State College, PA

Bachelor of Science in Computer Engineering, GPA: 3.94/4.0

Expected Graduation: May 2026

EXPERIENCE

AMD

Austin, TX

GPU Performance Intern (AI Models)

May 2026 – Present

- Designing an automated GPU profiling pipeline for the machine learning frameworks (ROCm kernels) team, overcoming current profiling-level limitations using vLLM and kineto traces for performance analysis.

AlphaGPU

Palo Alto, CA

Software Engineer Intern (Infrastructure)

Aug 2025 – Jan 2026

- Deployed GPU profiling and benchmarking for 1000+ users with NVML and torch profiler, overcoming cloud-hosted GPU limitations using NSYS/NCU for performance analysis.
- Built multi-file submission infrastructure linking 10+ .cu files and mounting them in Docker containers for compilation, debugging across a multi-language, multi-hardware system.
- Scaled LeetGPU, a platform for hardware-accelerated programming, exposing low-level GPU compute through CLI tools supporting CUDA, Triton, PyTorch, Mojo, and CuTe DSL.

NeuroAI Lab

State College, PA

AI Acceleration Research Assistant

Feb 2025 – Present

- Designed a novel, scalable high-dimensional Bayesian optimization algorithm using dynamic sampling for hyperparameter tuning across large search spaces.
- Researching hardware-software co-design for neuromorphic neural networks and novel architectures, applying system-level modeling to benchmark accuracy under device constraints.

nth Solutions

Coatesville, PA

Electrical Engineering & Data Analyst Intern

June 2023 – July 2024

- Developed and benchmarked 3 ML-driven signal processing algorithms, analyzing 100+ nonstationary time-series in Python and MATLAB to evaluate performance.
- Co-inventor on 2 products under patent preparation involving sensor-based safety and vibration reduction systems.

PROJECTS

FlashAttention CUDA Kernel | CUDA, C++, PyTorch

- Developed a FlashAttention-inspired CUDA kernel handling 8x more tokens than naive self-attention with 1.54x speedup, scaling to 200k+ context on a Tesla T4; benchmarked against PyTorch SDPA.
- Achieved O(N) memory via tiling and online softmax to minimize HBM accesses, and resolved shared-memory bank conflicts with strided padding plus FP16 tuning to maximize occupancy and throughput.

CUDA-Accelerated Neural Network from Scratch | CUDA, C++, Python, NumPy

- Built a deep neural network from scratch (no ML libraries) reaching 97.83% on MNIST and 88.02% on Fashion-MNIST with a custom dynamic learning-rate optimizer.
- Accelerated training 10.57x (40 to 3.8 minutes) with a custom CUDA matmul kernel exploiting GPU parallelization.

Essential Machine Learning Projects | Python, PyTorch, Transformers, Scikit-Learn

- Built a RAG (retrieval-augmented generation) bot with a TinyLlama LLM, a pneumonia-detection CNN reaching 90%+ accuracy, and a reinforcement learning agent, with data preprocessing pipelines.

TECHNICAL SKILLS

Languages: Java, CUDA, C/C++, Python, Go, Verilog/SystemVerilog, MIPS Assembly

ML & Algorithms: nearest-neighbor / vector search algorithms, clustering, dimensionality reduction, RAG (retrieval-augmented generation), PyTorch, Transformers, Scikit-Learn, NumPy, Bayesian optimization, linear algebra

GPU & HPC: CUDA kernels, GPU parallelization, memory hierarchy / HBM optimization, occupancy & throughput tuning, NVML, torch profiler, NSYS/NCU, CuTe DSL, Triton, Mojo

Core Competency Areas: performance analysis & benchmarking, accelerated computing, HPC, debugging multi-language / multi-hardware systems, AI/ML optimization, software development lifecycle, data analysis

Tools: Docker, Linux, LLVM, MATLAB CAPTAIN toolbox